

# 30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

## Coordinate Geometry

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

### Question 1

**Find the distance between A(-1, 2) and B(5, 10).**

Answer:

$$AB = \sqrt{[(5-(-1))]^2 + (10-2)^2} = 10 \text{ units.}$$

### Question 2

**Find the distance between A(0, 0) and B(6, 8).**

Answer:

$$AB = \sqrt{[(6-0)]^2 + (8-0)^2} = 10 \text{ units.}$$

### Question 3

**Find the distance between A(2, 3) and B(8, 11).**

Answer:

$$AB = \sqrt{[(8-2)]^2 + (11-3)^2} = 10 \text{ units.}$$

### Question 4

**Find the distance between A(-3, -4) and B(3, 4).**

Answer:

$$AB = \sqrt{[(3-(-3))]^2 + (4-(-4))^2} = 10 \text{ units.}$$

### Question 5

**Find the distance between A(1, 1) and B(4, 5).**

Answer:

$$AB = \sqrt{[(4-1)]^2 + (5-1)^2} = 5 \text{ units.}$$

### Question 6

**Find the distance between A(-2, 5) and B(4, 5).**

Answer:

$$AB = \sqrt{[(4-(-2))]^2 + (5-5)^2} = 6 \text{ units.}$$

### Question 7

**Find the distance between A(3, -1) and B(3, 8).**

Answer:

$$AB = \sqrt{[(3-3)]^2 + (8-(-1))^2} = 9 \text{ units.}$$

### Question 8

**Find the distance between A(0, 7) and B(24, 7).**

Answer:

$$AB = \sqrt{[(24-0)]^2 + (7-7)^2} = 24 \text{ units.}$$

### Question 9

**Find the point dividing A(2, -1) and B(8, 5) internally in ratio 1:2.**

Answer:

$$P = \left[ \frac{(1 \times 8 + 2 \times 2)}{3}, \frac{(1 \times 5 + 2 \times (-1))}{3} \right] = (4, 1).$$

## Coordinate Geometry - Continued

### Question 10

Find the point dividing A(0, 0) and B(9, 6) internally in ratio 1:2.

Answer:

$$P = [(1x_2+2x_1)/3, (1y_2+2y_1)/3] = (3, 2).$$

### Question 11

Find the point dividing A(-2, 4) and B(4, 10) internally in ratio 1:1.

Answer:

$$P = [(1x_2+1x_1)/2, (1y_2+1y_1)/2] = (1, 7).$$

### Question 12

Find the point dividing A(1, 3) and B(7, 9) internally in ratio 2:1.

Answer:

$$P = [(2x_2+1x_1)/3, (2y_2+1y_1)/3] = (5, 7).$$

### Question 13

Find the point dividing A(3, -3) and B(9, 3) internally in ratio 1:2.

Answer:

$$P = [(1x_2+2x_1)/3, (1y_2+2y_1)/3] = (5, -1).$$

### Question 14

Find the point dividing A(2, 5) and B(8, -1) internally in ratio 2:1.

Answer:

$$P = [(2x_2+1x_1)/3, (2y_2+1y_1)/3] = (6, 1).$$

### Question 15

Find the area of the triangle with vertices (0,0), (4,0) and (0,3).

Answer:

It is right-angled. Area =  $\frac{1}{2} \times 4 \times 3 = 6$  square units.

### Question 16

Find the area of the triangle with vertices (0,0), (6,0) and (0,5).

Answer:

It is right-angled. Area =  $\frac{1}{2} \times 6 \times 5 = 15$  square units.

### Question 17

Find the area of the triangle with vertices (0,0), (8,0) and (0,7).

Answer:

It is right-angled. Area =  $\frac{1}{2} \times 8 \times 7 = 28$  square units.

### Question 18

Find the area of the triangle with vertices (0,0), (10,0) and (0,4).

Answer:

It is right-angled. Area =  $\frac{1}{2} \times 10 \times 4 = 20$  square units.

### Question 19

Find the area of the triangle with vertices (0,0), (12,0) and (0,9).

Answer:

It is right-angled. Area =  $\frac{1}{2} \times 12 \times 9 = 54$  square units.

## Coordinate Geometry - Continued

### Question 20

**Find the area of the triangle with vertices (0,0), (14,0) and (0,6).**

Answer:

It is right-angled. Area =  $\frac{1}{2} \times 14 \times 6 = 42$  square units.

### Question 21

**State the main formula, rule or theorem used in Coordinate Geometry.**

Answer:

Distance =  $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ ; section formula uses weighted coordinates.

### Question 22

**What do the important symbols or quantities in Coordinate Geometry represent?**

Answer:

Coordinates locate a point by its x-value and y-value.

### Question 23

**Write the correct method for solving a Coordinate Geometry numerical.**

Answer:

Draw a rough diagram, write the appropriate formula, substitute coordinates with brackets and simplify.

### Question 24

**Which unit, sign convention or condition must be checked in a Coordinate Geometry answer?**

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

### Question 25

**Why must the final answer in Coordinate Geometry be checked against the given condition?**

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

### Question 26

**What is one common error in Coordinate Geometry questions?**

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

### Question 27

**How should full working be shown in a Coordinate Geometry numerical?**

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

### Question 28

**When is a diagram, table or graph useful in Coordinate Geometry?**

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

## Coordinate Geometry - Continued

### Question 29

**How can a student decide which formula to use in Coordinate Geometry?**

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

### Question 30

**What makes a high-scoring answer in Coordinate Geometry?**

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

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